

Ryo Wijaya

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EDUCATION

National University of Singapore Singapore	Aug. 2020 – June 2024
<i>Bachelor of Computing, Information Systems (Hons. w/ Distinction) FinTech Specialisation</i>	
Heidelberg International School Germany	Aug. 2016 – May 2018
<i>International Baccalaureate (IB) Diploma</i>	

EXPERIENCE

Software Engineer Crédit Agricole Corporate & Investment Bank	July 2024 – Present
- Contributed to the design and development of a greenfield project to build modular ETL Treatment Chains using Spark Streaming, Spring Boot, and GCP; developed cloud-enabled Java drivers to ingest, enrich, and distribute Commercial Real Estate Loans, ESG, and Capital Markets data. - Owner and main contributor (7 out of 9 core modules) of a Spring Boot DevTools platform for automated JSON ↔ Avro conversion, Kafka topic management, PostgreSQL configuration validation and loading, and Apache Iceberg DDL from Avro schema generation; accelerating cross-functional delivery for teams in Singapore and Paris. - Architected and implemented OpenID Connect (OIDC) and OAuth2 SSO authentication for the Angular application platform and developed UI features. - Implemented schema validation across upstream sources and downstream destinations by utilizing Confluent Schema Registry and Kafka. Managed schema evolution across Avro schema version upgrades. - Orchestrated deployments using Helm, Kubernetes, and created GitLab CI pipelines to ensure robust CI/CD. - Contributed to application security by remediating Checkmarx and JSFrog Xray vulnerabilities, and enforcing SonarQube quality gates.	
Software Engineer Intern SP Group	Jan. 2023 – June 2023
- Developed an OCPP 1.6/2.0.1 compliant electric vehicle charger simulator using FastAPI and WebSockets. - Led load testing for over 2,000 simulated chargers to identify infrastructure scalability bottlenecks. - Built a Next.js platform for real-time charger configuration, message triggering, and log monitoring.	
Software Engineer Intern Reluvate Technologies	May 2022 – Aug. 2022
- Developed a hiring web platform using Node and GCP Cloud Services to automate over 90% of company recruitment processes, including document exchange and technical assessments. - Built a Python backend for an accounting platform product to handle payments and contracts.	

SOFTWARE PROJECTS

FinRAG | *FastAPI, LangGraph, Qdrant, GCP* | Hybrid (vector + BM25) document search and analysis for financial data
PetHub | *Express.js, Next.js, PostgreSQL, Prisma, AWS* | Fully featured E-commerce pet-service marketplace
Optimal Travel Plan | *Jakarta EE, JSF, MySQL, Ionic* | Optimized travel itinerary generation app
my-pi-board | *Next.js* | Bus timings and stock market monitoring dashboard that lives on my Raspberry Pi
ryo | *React, Vite, Bootstrap* | Personal portfolio website
blog | *Jekyll, GitHub Actions* | Tech, finance, and hobby blog utilizing the open-source Chirpy theme
HotelReservationProject | *Jakarta EE, SOAP* | Enterprise hotel management and reservation system CLI
sprout | *Solidity, Truffle, Web3.js* | Blockchain job marketplace with DAO dispute resolution
ibkr-tv-tradebot | *FastAPI, IBKR TWS API* | Tradebot for order execution and TradingView integration

SKILLS

Languages: Java, Python, JavaScript/TypeScript, HTML, SQL
Technologies: Spring Boot, Apache Kafka (Confluent), Apache Spark, Apache Iceberg, FastAPI, Next.js, React.js, Node.js, Angular, PostgreSQL, NoSQL Databases, Jakarta EE, Jakarta Server Faces, OCPP
Infrastructure & Tools: Kubernetes, Helm, Docker, GitLab CI/CD, Git, Google Cloud Platform (Cloud Run, Artifact Registry, GKE, Compute Engine, Cloud Storage, Cloud SQL, etc), Azure (VM, AKS, Functions, Cosmos DB, Storage, etc)
AI Engineering: RAG, LangGraph, LangChain, Qdrant, Ollama
Interests: Electric Guitar, Piano, Drums, Music Production
Communication: English (Native), German (Conversational), Mandarin (Conversational)

CERTIFICATIONS

Certified Kubernetes Application Developer (CKAD) - 98/100 <i>Cloud Native Computing Foundation</i>	2025
Confluent Apache Kafka Fundamentals Accreditation <i>Confluent</i>	2026
Financial Technology Specialisation (Distinction) <i>National University of Singapore</i>	2024
IS Technology Knowledge Area (Distinction) <i>National University of Singapore</i>	2024